

ABDELRAHMAN ELDERAA

Computer Vision Engineer | Deep Learning | AI/ML

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PROFESSIONAL SUMMARY

Computer Vision Engineer with 1+ year of hands-on experience building and deploying deep learning models for medical imaging, real-time object detection, and semantic segmentation. Delivered production-grade CV pipelines — from raw data ingestion and augmentation through transfer learning, custom loss functions, model optimization, and Streamlit deployment. Achieved 87.5% Dice / 79.3% IoU on coronary artery segmentation and top classification accuracy using VGG19 transfer learning on MRI data. Additional expertise in LLM integration and Retrieval-Augmented Generation (RAG) systems using LangChain, FAISS, and Google Gemini.

EDUCATION

M.Sc. Data Science | *Cairo University* 2026 – Present

- First year — coursework phase covering advanced ML, statistical modeling, and data engineering.

B.Sc. Computer Science | *Modern Academy, Cairo* Oct 2020 – Jun 2024

- GPA: 3.57 / 4.0 — Top academic performance.
- ECPC (Egypt Collegiate Programming Contest) participant — competitive algorithmic problem-solving.

PROFESSIONAL EXPERIENCE

AI Engineer — Computer Vision | *Balance Code Company* Sep 2025 – Present

- Developed and optimized deep learning models for real-time image/video analysis using YOLO and Attention-based CNNs.
- Designed and fine-tuned U-Net architectures for semantic segmentation, improving pixel-wise accuracy on custom datasets.
- Built end-to-end CV pipelines: data ingestion, preprocessing, training, evaluation, and deployment.
- Benchmarked models using Precision, Recall, F1, IoU, and confusion matrices; iteratively refined architectures.

AI Instructor — ML & Computer Vision | *Mindset Company* Feb 2025 – Sep 2025

- Delivered hands-on workshops on image segmentation (U-Net), object detection, and regression using Python and TensorFlow.
- Designed project-based curriculum for building and evaluating real-world CV models.
- Mentored 30+ students through CV and ML capstone projects.

Supervisor & Program Coordinator | *Mindset Company* Jul 2024 – Jan 2025

- Managed AI course delivery, tracking student progress and resolving technical issues in real time.
- Reviewed assignments and final projects; provided structured feedback on Python and ML fundamentals.

Teaching Assistant — Computer Science | *Modern Academy* Aug 2024 – Aug 2025

- Conducted tutorials on algorithms, data structures, and programming for undergraduate students.
- Developed instructional materials and collaborated with faculty to improve course delivery.

KEY PROJECTS

Coronary Artery FFR Estimation System — *Python · PyTorch · UNet++ · OpenCV*

- Built a deep learning pipeline using UNet++ (VGG16-BN encoder) to segment coronary arteries frame-by-frame from angiography videos; trained on 9,867 pairs with custom loss (cIDice $\times 0.7$ + Focal Tversky $\times 0.3$) achieving Dice 87.50% / IoU 79.29%.
- Implemented QCA to compute Diameter Stenosis (DS%) from proximal/distal RVD and MLD extracted from segmentation masks; estimated FFR using Wibmer 2018 formula, classifying results into 4 severity levels.
- Developed multi-vessel tracker with IoU-based matching, bifurcation detection, loop exclusion, and 7-frame temporal smoothing; exported per-vessel clinical CSV reports with annotated video overlay.

Alzheimer's Detection — Multi-Model CNN Comparison — *Graduation Project · TensorFlow · Keras*

- Built and evaluated 3 CNN architectures (custom CNN, VGG16, VGG19) using transfer learning on MRI brain scan datasets; VGG19 achieved best classification accuracy, with results contributing to a published academic report.
- Applied data preprocessing and augmentation to mitigate overfitting; compared models with Precision, Recall, F1-score, and confusion matrices.

Medical Image Segmentation — U-Net with VGG19 Encoder — *TensorFlow · Keras · OpenCV*

- Developed a semantic segmentation pipeline using U-Net + pretrained VGG19 encoder for binary mask prediction on medical imaging datasets; evaluated with IoU and Dice coefficient metrics.
- Processed multi-format datasets using TensorFlow, Keras, and OpenCV; pipeline produced high-resolution segmentation masks that improved diagnostic precision.

Real-Time Face Mask Detection — *Python · YOLOv8 · OpenCV*

- Built a real-time object detection system using YOLOv8 to identify face mask compliance in public spaces; trained on custom-labeled dataset with Roboflow annotation pipeline.
- Achieved high precision and recall on live video streams; deployed inference pipeline with OpenCV for real-time visualization.

News Research Tool & RAG Chatbot — *LangChain · FAISS · Llama 3 · Gemini · Streamlit*

- Built two production RAG Q&A systems: news research tool using LangChain + Llama-3.3-70B + FAISS with automated scraping and chunking; knowledge-base chatbot using Google Gemini + HuggingFace embeddings.
- Both deployed via Streamlit with hallucination-controlled prompting; demonstrated end-to-end LLM pipeline design from ingestion to inference.

Plant Disease Detection — CNN Classifier — *TensorFlow · CNN · PlantVillage Dataset*

- Designed and trained a CNN using TensorFlow on the PlantVillage dataset to classify potato leaf diseases; trained for 50+ epochs with learning-rate scheduling and dropout regularization for improved robustness.

CERTIFICATIONS & TRAINING

- Huawei HCIA-AI V3.5 — NTI (Jun 2025 – Jul 2025) | AI fundamentals, ML models, Huawei AI solutions
- Artificial Intelligence Diploma — AMIT Learn (Jul 2023 – Mar 2024) | Python, Data Science, ML, Deep Learning, NLP, Computer Vision
- AI & Machine Learning Internship — Instant (Jun 2023 – Aug 2023) | Mathematics, Data Analysis, Data Science, Deep Learning

CORE TECHNICAL SKILLS

Computer Vision	OpenCV, CNNs, U-Net, YOLO (v5/v8/v11), UNet++, Transfer Learning, Attention Mechanisms, Image Segmentation, Object Detection
Deep Learning	TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost
Architectures	VGG16/VGG19, ResNet, U-Net with custom encoders, UNet++ (VGG16-BN), Siamese Networks, Attention CNNs
NLP & LLMs	LangChain, FAISS, Pinecone, RAG pipelines, Llama 3, Google Gemini, Cohere, NLTK, spaCy
Data & Viz	NumPy, Pandas, Matplotlib, Seaborn, Power BI
Deployment	Streamlit, FastAPI, Flask, Gradio, Roboflow, CVAT
Languages	Python (primary), C++, Java, SQL
Tools	Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Linux

LANGUAGES

- Arabic — Native proficiency
- English — Intermediate (reading, writing, technical communication)